

Complex Numbers



Powers of i

1 Evaluate each power of i , using $i^2 = -1$ and the fact that powers of i repeat every 4:

a i^3

b i^4

c i^7

d i^{22}

2 Evaluate i^{101} .

Addition and subtraction

3 Let $z_1 = 3 + 4i$ and $z_2 = -2 + 5i$. Find $z_1 + z_2$.

4 Let $z_1 = 2 - 3i$ and $z_2 = -1 - 4i$. Find $z_1 + z_2$.

5 Find $(7 + 2i) - (3 - 5i)$.

6 Find $(4 - i) - (-2 + 6i)$.

Multiplication

7 Multiply $(2 + 3i)(1 - 4i)$.

8 Multiply $(-1 + 2i)(3 + i)$.

9 Multiply $(5 - 2i)(5 + 2i)$, and explain why the result has no imaginary part.

Conjugate and modulus

10 For $z = 6 - 8i$, find $\bar{z}$ and $|z|$.

11 For $z = -5 + 12i$, find $\bar{z}$ and $|z|$.

12 Show that $z + \bar{z}$ is always real for $z = a + bi$.

Division

- 13 Simplify $\frac{3+4i}{1-2i}$ by multiplying top and bottom by the conjugate of the denominator.
- 14 Simplify $\frac{1+i}{1-i}$.
- 15 Simplify $\frac{7-i}{3+2i}$.
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Quadratic equations with complex roots

- 16 Solve $x^2 + 9 = 0$.
- 17 Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.
- 18 Solve $2x^2 + 2x + 5 = 0$ using the quadratic formula.
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Multiple choice — easy

- 19 Evaluate i^{130} , using the fact that powers of i repeat every 4.
- a. i
 - b. $-i$
 - c. 1
 - d. -1
- 20 Evaluate i^{160} , using the fact that powers of i repeat every 4.
- a. -1
 - b. $-i$
 - c. 1
 - d. i

21 Evaluate i^{211} , using the fact that powers of i repeat every 4.

a. 1

b. -1

c. $-i$

d. i

22 Evaluate i^{236} , using the fact that powers of i repeat every 4.

a. 1

b. -1

c. i

d. $-i$

23 Evaluate i^{229} , using the fact that powers of i repeat every 4.

a. 1

b. i

c. -1

d. $-i$

24 Evaluate i^{52} , using the fact that powers of i repeat every 4.

a. -1

b. 1

c. $-i$

d. i

25 Evaluate i^{206} , using the fact that powers of i repeat every 4.

a. i

b. $-i$

c. -1

d. 1

26 Evaluate i^{81} , using the fact that powers of i repeat every 4.

a. -1

b. $-i$

c. 1

d. i

27 Evaluate i^{62} , using the fact that powers of i repeat every 4.

a. i

b. -1

c. 1

d. $-i$

28 Evaluate i^{231} , using the fact that powers of i repeat every 4.

a. 1

b. -1

c. $-i$

d. i

29 Evaluate i^{101} , using the fact that powers of i repeat every 4.

a. $-i$

b. -1

c. i

d. 1

30 Evaluate i^{66} , using the fact that powers of i repeat every 4.

a. i

b. 1

c. $-i$

d. -1

31 Evaluate i^{31} , using the fact that powers of i repeat every 4.

a. -1

b. i

c. 1

d. $-i$

32 Evaluate i^7 , using the fact that powers of i repeat every 4.

a. $-i$

b. 1

c. i

d. -1

33 Evaluate i^{178} , using the fact that powers of i repeat every 4.

- a. i
- b. 1
- c. -1
- d. $-i$

Multiple choice — medium

34 Multiply $(-9 - 4i)(1 - 6i)$.

- a. $-33 + 15i$
- b. $-33 + 50i$
- c. $15 + 50i$
- d. $-33 + 58i$

35 Multiply $(3 - i)(-3 - 9i)$.

- a. $-18 - 24i$
- b. $-18 - 30i$
- c. $-24i$
- d. -18

36 Multiply $(-3 - 2i)(-3 - 2i)$.

- a. $5 + 13i$
- b. $5 + 12i$
- c. $13 + 12i$
- d. 5

- 37 Multiply $(7 + 2i)(4)$.
- a. $29 + 8i$
 - b. $28 + 8i$
 - c. $28 - 8i$
 - d. $28 + 28i$
- 38 Multiply $(-8 - 9i)(-5 - i)$.
- a. $49 + 53i$
 - b. $31 + 49i$
 - c. $31 + 53i$
 - d. $31 - 37i$
- 39 Multiply $(7 - 6i)(4 + 8i)$.
- a. $-20 + 32i$
 - b. $76 + 80i$
 - c. $76 - 20i$
 - d. $76 + 32i$
- 40 Multiply $(3 - 9i)(-8 + 3i)$.
- a. $3 - 51i$
 - b. $3 - 63i$
 - c. $3 + 81i$
 - d. $-51 + 81i$

41 Multiply $(-3 - 7i)(1 - 3i)$.

a. $-24 + 18i$

b. $-24 + 2i$

c. $18 + 2i$

d. $-24 + 16i$

42 Multiply $(3 + 6i)(5 + 6i)$.

a. $51 + 48i$

b. $-21 + 48i$

c. $-21 - 12i$

d. $-21 + 51i$

43 Multiply $(8 - 5i)(4 + 6i)$.

a. $62 + 2i$

b. $62 + 68i$

c. $62 + 28i$

d. $2 + 28i$

44 Multiply $(8 - 5i)(9 + 4i)$.

a. $92 + 52i$

b. $52 - 13i$

c. $92 - 13i$

d. $92 + 77i$

45 Multiply $(-i)(-3 + 8i)$.

a. $8 - 3i$

b. $8 - 8i$

c. $-8 + 3i$

d. $8 + 3i$

46 Multiply $(2 - 2i)(-6 - 5i)$.

a. $-22 - 2i$

b. $-22 + 2i$

c. $-2 + 2i$

d. $-22 - 22i$

47 Multiply $(-1 - 2i)(9 - 4i)$.

a. $-17 + 22i$

b. $-17 - i$

c. $-1 - 14i$

d. $-17 - 14i$

48 Multiply $(-7 - 6i)(2 + 3i)$.

a. $4 - 9i$

b. $4 - 33i$

c. $4 - 32i$

d. $-32 - 33i$

Multiple choice — hard

- 49 Simplify $\frac{-34 - 8i}{-7 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.
- a. $-12/53 + 254/53i$
 - b. $307/53 - 12/53i$
 - c. $254/53 - 12/53i$
 - d. $254/53 + 12/53i$
- 50 Simplify $\frac{6 + 8i}{2 + 8i}$ by multiplying top and bottom by the conjugate of the denominator.
- a. $-8/17 + 19/17i$
 - b. $19/17 + 8/17i$
 - c. $19/17 - 8/17i$
 - d. $36/17 - 8/17i$
- 51 Simplify $\frac{-10 - 54i}{-1 - 9i}$ by multiplying top and bottom by the conjugate of the denominator.
- a. $248/41 - 18/41i$
 - b. $289/41 - 18/41i$
 - c. $248/41 + 18/41i$
 - d. $-18/41 + 248/41i$
- 52 Simplify $\frac{16 + 4i}{4 + 2i}$ by multiplying top and bottom by the conjugate of the denominator.
- a. $18/5 - 4/5i$
 - b. $-4/5 + 18/5i$
 - c. $18/5 + 4/5i$
 - d. $23/5 - 4/5i$

- 53 Simplify $\frac{-45 - 12i}{-8 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.
- a. $3/34 + 96/17i$
 - b. $96/17 + 3/34i$
 - c. $96/17 - 3/34i$
 - d. $113/17 + 3/34i$
- 54 For $z = 9 - 40i$, find $|z|$.
- a. 49
 - b. 42
 - c. 40
 - d. 41
- 55 For $z = 3 - 4i$, find $|z|$.
- a. 5
 - b. 4
 - c. 6
 - d. 7
- 56 For $z = 5 + 12i$, find $|z|$.
- a. 17
 - b. 12
 - c. 14
 - d. 13

- 57 For $z = 7 + 24i$, find $|z|$.
- a. 26
 - b. 25
 - c. 24
 - d. 31
- 58 For $z = 8 - 15i$, find $|z|$.
- a. 17
 - b. 18
 - c. 16
 - d. 23
- 59 Solve $x^2 + 14x + 98 = 0$ using the quadratic formula.
- a. $-7 + 7i, -7 - 7i$
 - b. $-6 + 7i, -6 - 7i$
 - c. $-7 + 7i$
 - d. $7 + 7i, 7 - 7i$
- 60 Solve $x^2 - 14x + 113 = 0$ using the quadratic formula.
- a. $7 + 8i$
 - b. $-7 + 8i, -7 - 8i$
 - c. $8 + 8i, 8 - 8i$
 - d. $7 + 8i, 7 - 8i$

- 61 Solve $x^2 + 25 = 0$ using the quadratic formula.
- a. $5i, -5i$
 - b. $5i$
 - c. $1 + 5i, 1 - 5i$
 - d. $2 + 5i, 2 - 5i$
- 62 Solve $x^2 + 14x + 85 = 0$ using the quadratic formula.
- a. $-7 + 6i$
 - b. $-7 + 6i, -7 - 6i$
 - c. $-6 + 6i, -6 - 6i$
 - d. $7 + 6i, 7 - 6i$
- 63 Solve $x^2 + 16 = 0$ using the quadratic formula.
- a. $1 + 4i, 1 - 4i$
 - b. $4i$
 - c. $2 + 4i, 2 - 4i$
 - d. $4i, -4i$